



# Adaptive Chaotic Modulation for SatCom

Seoyoon (Yooni) Jung | [sj4260@nyu.edu](mailto:sj4260@nyu.edu)

Spring 2026 | ECE-UY 3404 Fundamentals of  
Communication Theory

# Constraint: SatCom

What is the env-specific constraint?

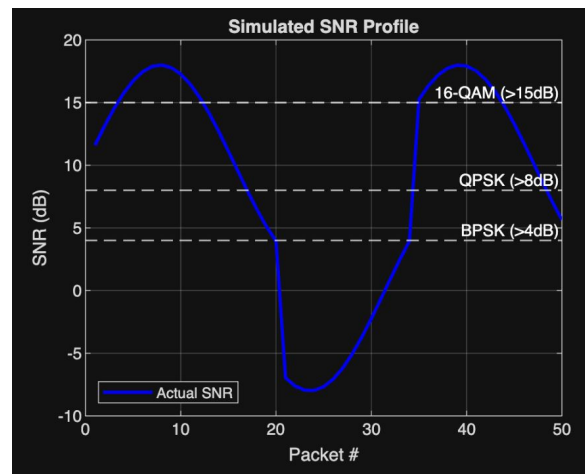
Rain, Scintillation (high-freq ionospheric disturbance), etc.

→ dynamic SNR

e.g. High-speed data (clear sky) / Standard telemetry /Critical health / Emergency or Deep fade

```
% Sim: Dynamic SNR
```

```
snr_profile = 10 + 8*sin(0.2*time) - (time > 20 & time <  
35)*10;
```



# Digital Modulation schemes

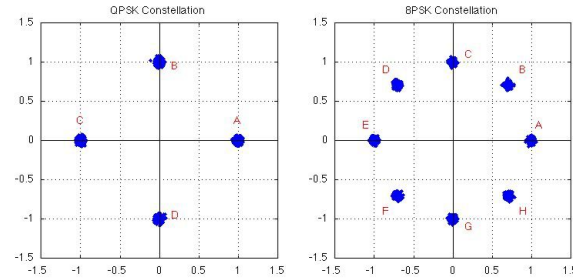
## DVB-S2 Standard

### DVB-S2: The Second Generation Standard for Satellite Broad-band Services

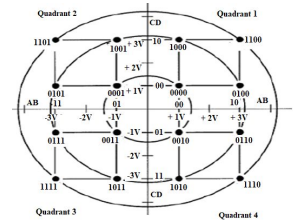
ALBERTO MORELLO AND VITTORIA MIGNONE

- **QPSK/ 8-PSK** for satellite links:  
Near-constant envelopes (for sat amplifiers)
- **16-APSK/ 32-APSK** for high SNR settings

### QPSK / 8PSK / 16-QAM constellation diagram



Source: [ece.unb.ca](http://ece.unb.ca)



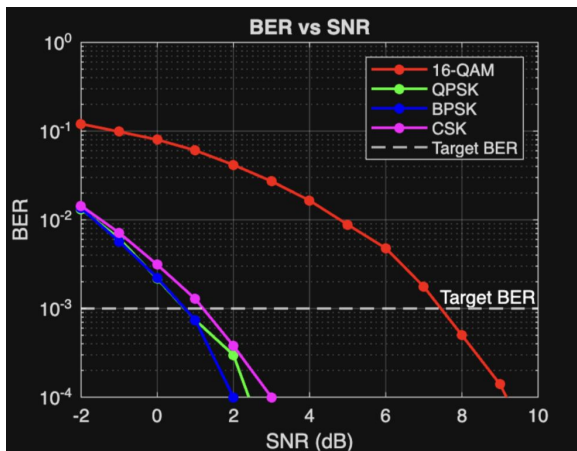
Source: [10.35940/ijrte.C5535.098319](https://doi.org/10.35940/ijrte.C5535.098319)

- Selected mod schemes in this project:

**16-QAM / QPSK / BPSK / + CSK**

# Performance comparison

Error rate in each scheme in noisy env



For image in low SNR (e.g. 2)

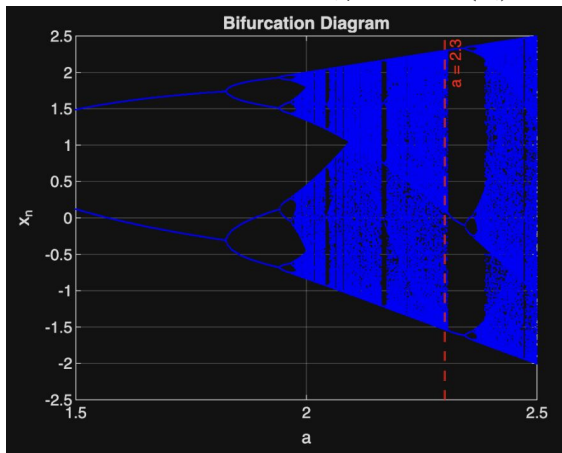


: Speed vs. Robustness Tradeoff

# CSK (Chaotic Shift Keying)

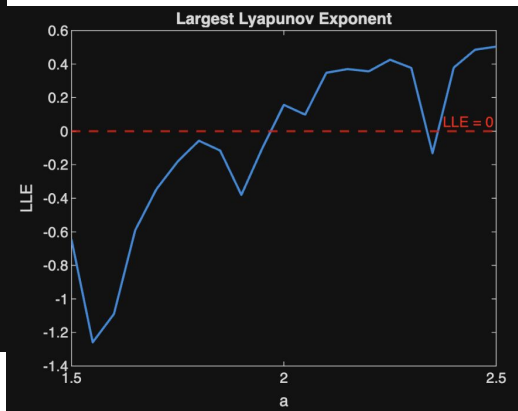
(adapted from Lab6)

$$x_{n+1} = a \cos(x_n)$$



$$\lambda = \frac{1}{N} \sum \log |f''(x_n)|$$

where for  $f(x) = a \cos(x)$



LLE > 0 means  
sensitive to initial conditions = chaotic system

LLE becomes positive for **a > 2**. → used a = 2.3



## CSK in SatCom

### Wireless Chaos-Based Communication Systems: A Comprehensive Survey

GEORGES KADDOUM, (Member, IEEE)

Chaos in Satellite-to-Ground links:  
**Security** without heavy computing

**CSK** as spread spectrum technique  
against jamming

# Adaptive Coding and Modulation

## ACM Controller

### Adaptive coding and modulation for satellite broadband networks: From theory to practice

[Hermann Bischl](#), [Hartmut Brandt](#), [Tomaso de Cola](#), [Riccardo De Gaudenzi](#) ✉, [Ernst Eberlein](#), [Nicolas Girault](#), [Eric Alberty](#), [Stefan Lipp](#), [Rita Rinaldo](#), [Bjarne Rislow](#) ... [See all authors](#) ▾

Switching logic: specific **target BER** policy

```
targetBER = 1e-3; % policy
```

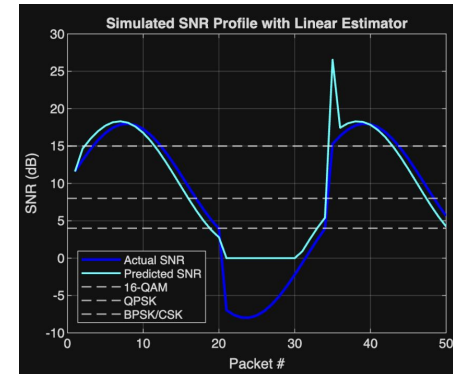


## Proactive estimator

### Channel Prediction and Adaptation over Satellite Channels with Weather-Induced Impairments by

Jihwan Patrick Choi

Make controller trend-aware



# Pipeline overview

## Signal

### 1) Input:

```
bits = randi([0 1], 1, bitsPerPacket); % TX: random bits
```

### 2) Channel:

```
channel_modes = {'awgn', 'awgn+interference'};  
sir_dB= 10; % interference(10dB below signal power)
```

### 3) SNR estimator (linear):

```
trend = currentSNR - snr_profile(t-1); % slope  
predictedSNR = currentSNR + trend; % extrapolate
```

### 4) ACM transceiver: linear (QAM/QPSK/BPSK) / chaotic (CSK)

```
[ber, mode, bitsPerSym] = adaptive_controller(bits, currentSNR, predictedSNR, nov, a, channel_mode, sir_dB);
```

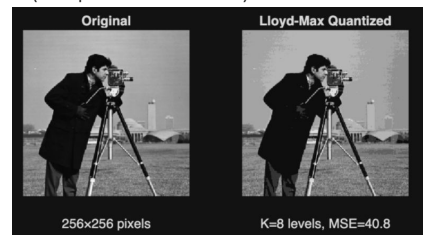
--> Performance metrics: BER vs SNR, BER vs Throughput, Reconstructed image, RMSE comparison table

## Image

### 1) Input -> Quantize (Lloyd-Max):

```
% e.g. [ 143 -> quantize -> 140 -> index -> 4 -> encode -> 011 ->  
transmit -> 011 -> decode -> 4 -> lookup -> 140 ]
```

(adapted from Lab6)

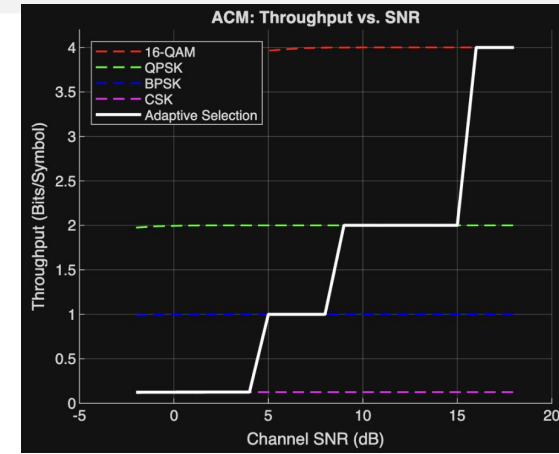


# Adaptive controller

```
function [ber, mode, bitsPerSym, rx_bits] = adaptive_controller( ...
    bits, currentSNR, predictedSNR, nov, a_chaos, channel_mode, sir_dB)

% Interference shifts all thresholds up: bias toward CSK earlier
if strcmp(channel_mode, 'awgn+interference')
    th = [18, 12, 6]; % tighter thresholds under jamming
else
    th = [15, 8, 4]; % standard thresholds
end

% High-speed data (clear sky)
if predictedSNR > th(1)
    [ber, rx_bits] = mod_qam(bits, currentSNR, 16, nov, channel_mode, sir_dB);
    mode = "16-QAM"; bitsPerSym = 4;
% Standard telemetry
elseif predictedSNR > th(2)
    [ber, rx_bits] = mod_qam(bits, currentSNR, 4, nov, channel_mode, sir_dB);
    mode = "QPSK"; bitsPerSym = 2;
% Critical health
elseif predictedSNR > th(3)
    [ber, rx_bits] = mod_qam(bits, currentSNR, 2, nov, channel_mode, sir_dB);
    mode = "BPSK"; bitsPerSym = 1;
% Emergency/Deep fade
else
    [ber, rx_bits] = mod_csk(bits, currentSNR, nov, a_chaos, channel_mode, sir_dB);
    mode = "CSK"; bitsPerSym = 1/nov;
end
end
```



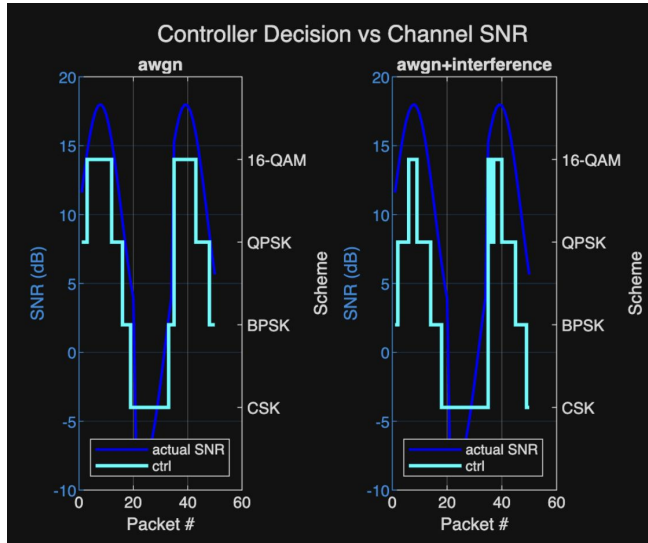
Supporting functions:

```
function [ber, rx_bits] = mod_qam(bits, snr_db, M, nov, channel, sir_db)
% bits: input binary vector
% snr_db: channel SNR in dB
% M: modulation order (2= BPSK, 4= QPSK, 16= 16-QAM)
% nov: oversampling rate (samples per symbol)
% channel: 'awgn' or 'awgn+interference'
% sir_db: signal-to-interference (dB)
```

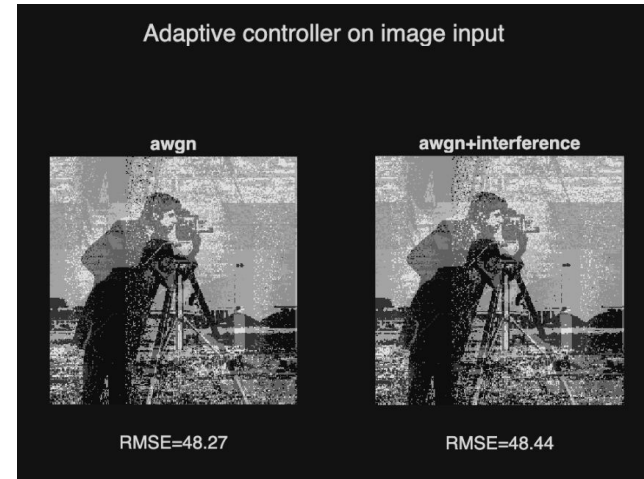
```
function [ber, rx_bits] = mod_csk(bits, snr_db, nov, a, channel, sir_db)
% bits: input binary vector
% snr_db: channel SNR in dB
% K: spreading factor (samples per bit); match nov in QAM
% a: Chaos parameter (>2)
% channel: 'awgn' or 'awgn+interference'
% sir_db: signal-to-interference (dB)
```

# Performance

For both channels,  
controller switches mod scheme according to SNR



RMSE evaluates image quality;  
i.e. Transmission time



# Future direction

- Determining threshold: closed-loop
- Added performance metrics
- Noncoherent Orthogonal BFSK (NBFSK) /  
Noncoherent Differential BPSK (DBPSK) in deep fade  
(as discussed on Apr 27)

# Reference

Morello, A., & Mignone, V. (2006). *"DVB-S2: The Second Generation Standard for Satellite Broad-band Services"*

Kaddoum, G. (2016). *"Wireless Chaos-Based Communication Systems: A Comprehensive Survey"*

Bischi, H., Brandt, H., de Cola, T., De Gaudenzi, R., Eberlein, E., Girault, N., Albery, E., Lipp, S., Rinaldo, R., Risløw, B., Skard, J.A., Tusch, J. and Ulbricht, G. (2010), *Adaptive coding and modulation for satellite broadband networks: From theory to practice. Int. J. Satell. Commun. Network.*, 28: 59-111. <https://doi.org/10.1002/sat.932>

Choi, J. (2000). *"Channel Prediction and Adaptation over Satellite Channels with Weather-Induced Impairments"*



# Adaptive Chaotic Modulation for SatCom

Seoyoon (Yooni) Jung [sj4260@nyu.edu](mailto:sj4260@nyu.edu)

ECE-UY 3404 Fundamentals of Communication Theory  
Spring 2026